

1. Let

$$f, g \in F[t], \quad g \neq 0.$$

Prove that there exist unique

$$q, r \in F[t]$$

such that

$$f = qg + r$$

and either

$$r = 0$$

or

$$\deg r < \deg g.$$

Your proof must establish both:

existence

and

uniqueness.

Do not assume polynomial long division as an algorithmic fact; derive the theorem from the field structure and degree.

2. Let

$$f, g \in F[t]$$

be not both zero. Prove that there exists a unique monic polynomial d such that

$$d \mid f, \quad d \mid g,$$

and

$$c \mid f, \ c \mid g \implies c \mid d.$$

Then prove that there exist

$$x, y \in F[t]$$

such that

$$\boxed{d = xf + yg.}$$

Your argument should ultimately depend on Problem 1. As part of your proof, justify why the Euclidean algorithm must terminate.

3. Let

$$\pi \in F[t]$$

be irreducible.

(a) Prove that if

$$\pi \nmid f,$$

then

$$\gcd(\pi, f) = 1.$$

(b) Using Problem 2, prove that

$$\pi \mid fg$$

implies

$$\pi \mid f$$

or

$$\pi \mid g.$$

Thus determine the relation between irreducibility and primality in $F[t]$.

(c) Prove that every nonzero nonunit polynomial in $F[t]$ can be written as a finite product of irreducible polynomials.

(d) Prove uniqueness: if

$$f = u\pi_1 \cdots \pi_r = v\rho_1 \cdots \rho_s,$$

where

$$u, v \in F^\times$$

and all π_i, ρ_j are irreducible, prove that

$$r = s$$

and, after reordering, each π_i is associate to ρ_i .

Conclude that every monic nonconstant polynomial has a unique factorization into monic irreducible polynomials, up to ordering.

4. Let

$$m \in F[t], \quad \deg m \geq 1.$$

Prove the equivalence

$$\boxed{F[t]/(m) \text{ is a field} \iff m \text{ is irreducible.}}$$

You must prove both directions. For the irreducible direction, begin with an arbitrary nonzero class

$$[f] \neq [0].$$

For the reducible direction, reason from a nontrivial factorization

$$m = ab.$$

Your proof should use results established earlier in today's sheet rather than treating this as an independent theorem.

5. Let

$$p$$

be prime and let

$$m \in \mathbb{F}_p[t]$$

be irreducible of degree

$$d \geq 1.$$

Set

$$K = \mathbb{F}_p[t]/(m).$$

(a) Prove that every element of K has a unique representative

$$a_0 + a_1t + \cdots + a_{d-1}t^{d-1}$$

with

$$a_i \in \mathbb{F}_p.$$

(b) Deduce that

$$\boxed{|K| = p^d.}$$

(c) Let

$$\alpha = [t] \in K.$$

If

$$m(t) = t^d + c_{d-1}t^{d-1} + \cdots + c_0,$$

prove that

$$m(\alpha) = 0$$

inside K . In other words,

$$\alpha^d + c_{d-1}\alpha^{d-1} + \cdots + c_0 = 0.$$

(d) Prove that every element of K can be written uniquely as

$$a_0 + a_1\alpha + \cdots + a_{d-1}\alpha^{d-1}.$$

(e) Explain precisely why K contains a copy of $\mathbb{F}_p$, but when $d > 1$,

$$K \neq \mathbb{F}_p.$$

At this point you will have constructed, from first principles, a field with

$$p^d$$

elements.